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ECON7335: Problem Set 1

1. **Economy 1 with endogenous information**: Consider a static economy in which agents' goal is to track an exogenous fundamental θ. That is, each agent's action is given by
$$a_i = E^i[\theta]. \tag{1}$$
Each agent's information set, Ω^i, consists of an exogenous, idiosyncratic signal
$$s_i = \theta + v_i \tag{2}$$
and an endogenous, aggregate signal
$$g = a + \eta. \tag{3}$$
The noise terms are distributed normally, with mean zero and variances σ_v^2 and σ_η^2 respectively. Moreover, the shocks v_i are uncorrelated across agents. Agents hold a flat (diffuse) prior over θ.

   (a) Conjecture an equilibrium relationship for aggregate action (call it the PLM)
   $$a = \theta + \phi_2 \eta. \tag{4}$$
   Derive the optimal inference of agent i given the conjecture. (Note that this conjecture restricts the coefficient on θ to be exactly one, rather than leaving it free; your answer to (b) should confirm that this restriction is consistent with equilibrium.)

   (b) Using your answer in (a), solve for the equilibrium law of motion (the ALM) of the aggregate action, given the conjectured PLM.

   (c) Write the fixed-point equation for ϕ_2 implied by part (b) above. Show that this relationship implies that the equilibrium coefficient ϕ_2 is the solution to a third-order equation. (But DON'T solve for it!)

   (d) Without solving the cubic, compute the limits of ϕ_2 as $\sigma_\eta \to 0$ and as $\sigma_\eta \to \infty$, and interpret each. In each limit, what happens to the variance of the noise term in the aggregate action a? Is that variance monotone in σ_η?

2. **Economy 2 with endogenous information:** Consider a static economy in which each agent is a mean-variance investor with unit risk aversion, holding a claim to the exogenous fundamental θ. That is, agent i chooses a_i to maximize $a_i E^i[\theta] - \frac{1}{2} a_i^2 Var(\theta|\Omega^i)$, so that

$$a_i = E^i[\theta]/Var(\theta|\Omega^i). \tag{5}$$

Each agent's information set, Ω^i, consists of an exogenous, idiosyncratic signal

$$s_i = \theta + v_i \tag{6}$$

and an endogenous, aggregate signal

$$g = a + \eta. \tag{7}$$

The noise terms are distributed normally, with mean zero and variances σ_v^2 and σ_η^2 respectively. Moreover, the shocks v_i are uncorrelated across agents. Agents hold a flat (diffuse) prior over θ. (The only change from Economy 1 is the rescaling of the optimal action: agents now respond more aggressively the more confident they are. You may read g as a market price and η as noise supply.)

(a) Conjecture an equilibrium relationship for aggregate action (call it the PLM)

$$a = \phi_1 \theta + \phi_2 \eta. \tag{8}$$

Derive the optimal inference of agent i given the conjecture.

(b) Using your answer in (a), solve for the equilibrium law of motion (the ALM) of the aggregate action, given the conjectured PLM.

(c) Write the fixed-point equation for ϕ_1 and ϕ_2 implied by part (b) above. Show that this relationship implies that the equilibrium coefficients are the solution to a linear equation. (And, please do solve for them!)

(d) Can you provide any intuition for the different form of the results in 1.c and 2.c?